



STINK3114 NATURAL LANGUAGE PROCESSING

Semester II Session 2025/2026 (A252)

Assignment 1: Literature Review

Using Language Models for assessment of users' satisfaction with their partner in Malaysia

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Related Work

1. Discussion of Past Studies Thematic.

The recent research in the field of Natural Language Processing (NLP) and sentiment analysis can be logically divided into three themes.

- **Traditional Machine Learning on Social Media Sentiment:** Early works often used traditional machine learning to measure the sentiment of the people. As a case study, Support Vector Machines (SVM), Naive Bayes, and Logistic Regression, have all been used to analyse Twitter data in Malaysia in relation to mental health awareness as well as to identify offensive Malay internet slang.
- **In Deep Learning and Code-Mixed Challenges:** The second theme is the linguistic complexity of the Malaysian digital space, namely, code-mixing (mixing Malay and English). Nowadays, the research indicates that old-fashioned techniques are futile in this case, and deep learning should be employed. Scholars have managed to use Multilingual Bidirectional Encoder Representations from Transformer (M-bERT) in order to analyse mixed-Malay-English datasets.
- **Evaluation Relationship Satisfaction through Language Models:** The third theme is the leading edge of NLP, which is concerned with highly personal relationships, as opposed to social phenomena. The use of pre-trained language models to evaluate the satisfaction of users with their partners has been ground breaking in the research based on Persian Twitter data. Moreover, the 2025 and 2026 studies have shown that Large Language Models (LLMs) are able to identify subtle relationship dynamics, emotional intimacy, and a profound conversational intent.

2. Methodological and Approach Comparison A critical comparison of these research works indicates that there is a change in the methods and approaches and performance is different. The classical lexicon-based and machine learning methods (such as SVM and Random Forest) are computationally efficient and work reasonably well on clean and formal data. Nonetheless, these procedures work on a strict word-based analysis. They have a very serious flaw: they often misunderstand the contextual specifics of the sentences code-switched, spelling variations, and informal grammar.

Conversely, deep learning approaches, particularly transformer-based models like M-BERT and advanced LLMs, show vastly superior performance when dealing with unstructured social media text. Whereas traditional models do not understand context, LLMs are able to capture the general feeling of complex paragraphs and are also capable of dealing with negations and intensifiers. However, one of the most interesting assumptions of transfer learning approaches (such as M-bERT) is that the pre-trained multilingual knowledge is

a perfect fit to the local dialects, which may still cause certain errors in the processing of more localised state-level Malay slang.

3. **Shortcomings of Existing Work** In spite of the achievements of deep learning, there are still serious limitations of the available literature. Most of the sentiment analysis studies conducted in the Malay language are very narrow and confined within the realms of the general or the masses, including COVID-19, mental health awareness, and politics. A very strong lack of attention to highly personal, intimate area themes.

Although Habibzadeh and Asadpour (2025) were able to demonstrate that the use of language models could classify relationship satisfaction and domestic problems, they were only able to use the Persian language as their data. The specific culture, idioms and code-mixed Bahasa Rojak that dominate relationships in the home setting of Malaysian households cannot be explained by models trained on either Persian or formal English. As a result, the current models lack the ability to measure intimate relationship sentiment in the Malaysian population appropriately.

4. Existent Research Gap that resulted in this study.

These constraints reveal a major research gap: a lack of an NLP model that would quantify the relationship satisfaction in a code-mixed Malay-English environment. Although existing literature confirms that M-bert is more effective in processing mixed Malaysian text using languages and that language models can be subsequently used to reliably estimate romantic partner satisfaction in low-resource languages, no article has combined the two methods.

The gap that is dealt with in this project is where the advanced language models can be used to determine the satisfaction of the users towards their partners in Malaysia. In such a way, it takes Malay NLP research out of the generic topic of the wider public and offers a highly contextualised, computationally efficient alternative to the traditional, resource-intensive relationship surveys.

References (APA 7th Edition)

1. The sentiment analysis on mental health awareness by Non-Governmental organisation's Twitter. (n.d.). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/10033345>
2. International Journal of Computer Science and Network Security (IJCSNS). (n.d.). Slangs and Short Forms of Malay Twitter Sentiment Analysis using Supervised Machine Learning - UNIVERSITI TEKNIKAL MALAYSIA MELAKA INSTITUTIONAL REPOSITORY. <https://eprints.utm.edu.my/id/eprint/25548/>
3. Cam, H., Cam, A. V., Demirel, U., & Ahmed, S. (2023, December 17). Sentiment analysis of financial Twitter posts on Twitter with the machine learning classifiers. <https://centaur.reading.ac.uk/114845/>
4. Md-Nor, S., et al. (2025). Semi-automatic sentiment identification for Malay-English code-switched data. *Journal of Advanced Research in Applied Sciences and Engineering Technology*. <https://ir.unimas.my/id/eprint/47390/3/5640-Article%20Text-28397-1-10-20250109.pdf>
5. Kong, J. T. H., Juwono, F. H., Ngu, I. Y., Nugraha, I. G. D., Maraden, Y., & Wong, W. K. (2023). A mixed Malay–English language COVID-19 Twitter Dataset: A sentiment analysis. *Big Data and Cognitive Computing*, 7(2), 61. <https://doi.org/10.3390/bdcc7020061>
6. Habibzadeh, Z., & Asadpour, M. (2025). Using language models for assessment of users' satisfaction with their partner in Persian. *Proceedings of the First Workshop on Language Models for Low-Resource Languages*, 78-88. <https://aclanthology.org/2025.loreslm-1.5.pdf>
7. Masterson, A., et al. (2026). The imitation of intimacy: Comparing satisfaction in intimate human and AI companion relationships. *HRI Companion '26*. https://www.researchgate.net/publication/401445166_The_Imitation_of_Intimacy_Comparing_Satisfaction_in_Intimate_Human_and_AI_Companion_Relationships
8. Castro, E., Calvo, H., & Kolesnikova, O. (2025). Emotion and intention detection in a large language model. *Mathematics*, 13(23), 3768. <https://doi.org/10.3390/math13233768>